ASSIGNMENT - 2

1. What is knowledge representation? Explain its importance.
2. Compare propositional logic and predicate logic.
3. Represent the following in predicate logic:
    a. All humans are mortal. Socrates is a human.
    b. Somebody loves pizza.
    c. If a person is a student, then they have a valid ID.
    d. There exists a person who knows how to play the piano.
    e. There is at least one person who speaks three languages.
4. Explain the ISA hierarchy and give an example.
5. What is frame notation in AI?
6. Describe how resolution works in first-order logic.
7. Define computable functions and predicates.
8. What is non-monotonic reasoning? How does it differ from monotonic reasoning?
9. Explain forward and backward chaining with an example.
10. What are Bayesian networks? Give a simple application example.
11. Define and explain Hidden Markov Models.
12. How does the truth maintenance system work in AI?
13. State and explain Bayes' Theorem with an example.
14. What is the role of utility functions in decision-making?
15. Explain the concept of semantic nets.